



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2021 Year 12 Course Assessment Task 4 (Trial Examination)

Monday 30 August, 2021

General instructions

- Working time – 3 hours.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 13)

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MXX.1 – Mr Sekaran

☐ 12MXX.2 – Ms Ham

Marker's use only.

QUESTION	1-10	11	12	13	14	15	16	%
MARKS	$\overline{10}$	$\overline{17}$	$\overline{16}$	$\overline{18}$	$\overline{12}$	$\overline{14}$	$\overline{13}$	$\overline{100}$

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided (labelled as page 13).

Questions

Marks

1. Consider the following statement.

1

$$x > 3 \text{ or } x < -3$$

Which of the following is the negation of the statement above?

(A) $x < 3 \text{ or } x > -3$

(B) $x < 3 \text{ and } x > -3$

(C) $x \leq 3 \text{ or } x \geq -3$

(D) $x \leq 3 \text{ and } x \geq -3$

2. Which of the following is a sixth root of i ?

1

(A) $e^{i\frac{\pi}{6}}$

(B) $e^{i\frac{\pi}{4}}$

(C) $e^{i\frac{3\pi}{4}}$

(D) $e^{i\frac{\pi}{12}}$

3. For a certain complex number z where $\arg(z) = \frac{\pi}{5}$, which of the following does $\arg(z^7)$ equal to?

1

(A) $-\frac{7\pi}{5}$

(B) $-\frac{3\pi}{5}$

(C) $\frac{2\pi}{5}$

(D) $\frac{3\pi}{5}$

4. A particle of mass m is moving in a straight line with the following force acting on it:

1

$$F = \frac{m}{x^3}(6 - 10x)$$

Which of the following is an expression for its velocity in any position x , if the particle starts from rest at $x = 1$?

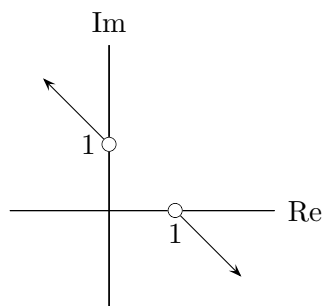
(A) $v = \pm \frac{1}{x} \sqrt{-3 + 10x - 7x^2}$

(C) $v = \pm \frac{\sqrt{2}}{x} \sqrt{-3 + 10x - 7x^2}$

(B) $v = \pm \sqrt{2}x \sqrt{-3 + 10x - 7x^2}$

(D) $v = \pm \frac{\sqrt{2}}{x} \sqrt{-3 + 10x + 7x^2}$

- 5.** The path traced out by a complex number z is shown on the Argand diagram below.



Which of the following is the equation of the path traced by z ?

- (A) $\arg(z - i) - \arg(z - 1) = 0$ (C) $\arg(z + i) - \arg(z + 1) = 0$
 (B) $\arg(z - i) - \arg(z - 1) = \pi$ (D) $\arg(z - i) - \arg(z - 1) = -\pi$

- 6.** Using the substitution $x = \pi - y$, which of the following will the definite integral **1**

$$\int_0^\pi x \sin x \, dx$$

simplify to?

- (A) 0

(B) $\int_0^\pi \sin x \, dx$

(C) $\frac{\pi}{2} \int_0^\pi \sin x \, dx$

(D) $\frac{\pi^2}{4}$

7. The vector $\underline{y}$ is given by $\underline{y} = \frac{1}{2} \begin{pmatrix} \sqrt{2} \\ -1 \\ 1 \end{pmatrix}$. Which of the following is the correct description of $\underline{y}$? 1

- (A) $\underline{y}$ makes an angle of 135° with the positive x -axis and 150° with the positive y -axis.
- (B) $\underline{y}$ makes an angle of 45° with the positive x -axis and 150° with the positive y -axis.
- (C) $\underline{y}$ makes an angle of 45° with the positive x -axis and 120° with the positive y -axis.
- (D) $\underline{y}$ makes an angle of 120° with the positive y -axis and 30° with the positive z -axis.

8. Given that $x, y \in \mathbb{Z}$, where $x, y \geq 0$, which of the following is a **FALSE** statement? 1

(A) $\forall x (\exists y : y = x)$

(C) $\forall x (\exists y : y = 1 + 2x)$

(B) $\exists x (\exists y : y = 2 - x)$

(D) $\exists x (\forall y : y = 1 - 2x)$

9. A particle is undergoing simple harmonic motion about a fixed point O . At time t seconds it has displacement x metres from O given by $x = a \cos nt$ for some constants $a > 0$ and $n > 0$. The period of the motion is T seconds. 1

What is the time taken by the particle to move from its starting position to a point half-way towards O ?

(A) $\frac{T}{12}$

(B) $\frac{T}{9}$

(C) $\frac{T}{8}$

(D) $\frac{T}{6}$

10. Which of these inequalities is **FALSE**? (Do NOT attempt to evaluate the integrals) 1

(A) $\int_1^2 \frac{1}{1+x} dx < \int_1^2 \frac{1}{x} dx$

(C) $\int_1^2 e^{-x^2} dx < \int_0^1 e^{-x^2} dx$

(B) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x}{x} dx < \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{x} dx$

(D) $\int_0^{\frac{\pi}{4}} \tan^2 x dx < \int_0^{\frac{\pi}{4}} \tan^3 x dx$

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (17 Marks)	Commence a NEW booklet.	Marks
(a) i. Prove that $\forall p \in \mathbb{Z}^+$,	If p^3 is even then p is even.	2
ii. Prove that $\sqrt[3]{2}$ is irrational.		3
(b) Suppose that n and $n + 1$ are positive integers, neither of which is divisible by 3.		3
Prove that $n^3 + (n + 1)^3$ is divisible by 9.		
(c) The sequence x_n is given by		
$x_1 = 1 \text{ and } x_{n+1} = \frac{4 + x_n}{1 + x_n} \text{ for } n \in \mathbb{Z}^+ \text{ where } n \geq 1$		
i. Prove by mathematical induction that for $n \in \mathbb{Z}^+$ where $n \geq 1$,		4
$x_n = 2 \left(\frac{1 + \alpha^n}{1 - \alpha^n} \right)$		
where $\alpha = -\frac{1}{3}$		
ii. Hence find the limiting value of x_n as $n \rightarrow \infty$.		1
(d) It is given that $a, b \in \mathbb{R}^+$.		
i. If $a + b = 6$, show	$\frac{1}{a} + \frac{1}{b} \geq \frac{2}{3}$	2
ii. If $a + b = c$, show	$\frac{1}{a^2} + \frac{1}{b^2} \geq \frac{8}{c^2}$	2

Examination continues overleaf...

Question 12 (16 Marks)

Commence a NEW booklet.

Marks

- (a) The points $A(1, -2, 3)$ and $B(-5, 4, -1)$ lie on the line ℓ_1 .

i. Show that a vector equation of ℓ_1 is $\underline{r} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 3 \\ -2 \end{pmatrix}$, where $\lambda \in \mathbb{R}$. **1**

- ii. Consider a line ℓ_2 with parametric equations **3**

$$\begin{cases} x = 1 - \mu \\ y = 2 + 3\mu \\ z = -1 + \mu \end{cases} \quad \text{where } \mu \in \mathbb{R}$$

Assuming ℓ_2 is neither parallel nor perpendicular to ℓ_1 , determine whether ℓ_1 and ℓ_2 intersect or are skew.

- (b) A sphere S_1 with centre $C(-3, -5, 10)$ passes through the point with coordinates $A(3, -3, 6)$.

- i. Show that the vector equation of S_1 is **1**

$$\left| \underline{u} - \begin{pmatrix} -3 \\ -5 \\ 10 \end{pmatrix} \right| = 2\sqrt{14}$$

- ii. Write down the Cartesian equation of S_1 . **1**

- iii. The vector equation of another sphere S_2 is **2**

$$\left| \underline{r} - \begin{pmatrix} -9 \\ 4 \\ 7 \end{pmatrix} \right| = \sqrt{14}$$

Prove that the two spheres S_1 and S_2 touch each other at a single point.

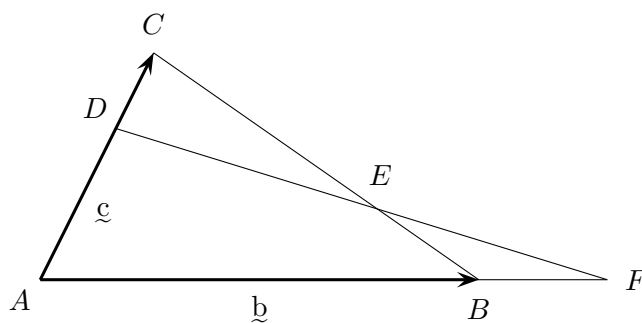
- iv. The vector equation of the line m is given as **3**

$$\underline{s} = \begin{pmatrix} -6 \\ -3 \\ 11 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \quad \text{where } \lambda \in \mathbb{R}$$

Find the value(s) of λ if the line m intersects the sphere S_1 twice.

Examination continues overleaf...

- (c) In $\triangle ABC$ below, D is the point on AC such that $AD : DC = 2 : 1$. E is the point on BC such that $BE : EC = 1 : 2$.



When DE is extended, it meets the extension of AB at F . Let $\overrightarrow{AB} = \mathbf{b}$ and $\overrightarrow{AC} = \mathbf{c}$.

- i. Show that $\overrightarrow{DE} = \frac{2}{3}\mathbf{b} - \frac{1}{3}\mathbf{c}$. **2**
- ii. Show that $AF : BF = 4 : 1$ **3**

[*Hint:* You may assume that $\overrightarrow{DF}$ is a scalar multiple of $\overrightarrow{DE}$, and $\overrightarrow{AF}$ is a scalar multiple of $\overrightarrow{AB}$]

Examination continues overleaf...

Question 13 (18 Marks)

Commence a NEW booklet.

Marks

- (a) It is given that $z = 1 + i$ is a root of the equation $z^3 + pz^2 + qz + 6 = 0$, where p and q are real. **3**

Find the value of p and q .

- (b) i. Using De Moivre's theorem, or otherwise, show that for every positive integer n , **3**

$$(1 + i)^n + (1 - i)^n = (\sqrt{2})^{n+2} \cos \frac{n\pi}{4}$$

- ii. Hence, or otherwise, show that for every positive integer n divisible by 4, **3**

$$\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots + \binom{n}{n} = (-1)^{\frac{n}{4}} (\sqrt{2})^n$$

Note: $\binom{n}{r} = {}^nC_k$

- (c) Let $\mu = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$. It is given that the complex number $\alpha = \mu + \mu^2 + \mu^4$ is a root of the quadratic equation $x^2 + ax + b = 0$, where $a, b \in \mathbb{R}$.

- i. Prove that $1 + \mu + \mu^2 + \dots + \mu^6 = 0$ **1**

- ii. The second root of the quadratic equation is β . Show with full working that **2**

$$\beta = \mu^3 + \mu^5 + \mu^6$$

- iii. Find the values of the coefficients a and b . **3**

- iv. Deduce that **3**

$$-\sin \frac{\pi}{7} + \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{2}$$

Examination continues overleaf...

Question 14 (12 Marks)

Commence a NEW booklet.

Marks

(a) Find

3

$$\int \frac{4x+3}{(x^2+1)(x+2)} dx$$

(b) i. Given $n \in \mathbb{Z}^+$, show that**1**

$$\sec^{2n} \theta = \sum_{k=0}^n \binom{n}{k} \tan^{2k} \theta$$

ii. Hence find

2

$$\int \sec^8 \theta d\theta$$

[Hint: Write $\sec^8 \theta$ as $\sec^6 \theta \sec^2 \theta$]

(c) i. By using a suitable substitution, or otherwise, evaluate

4

$$\int_0^{\frac{\sqrt{3}}{2}} \frac{x^3}{\sqrt{1-x^2}} dx$$

ii. Hence using integration by parts, or otherwise, evaluate

2

$$\int_0^{\frac{\sqrt{3}}{2}} 3x^2 \cos^{-1} x dx$$

Examination continues overleaf...

Question 15 (14 Marks)

Commence a NEW booklet.

Marks

- (a) i. Find **2**

$$\int \ln(1+x) dx$$

- ii. Let $I_n = \int_0^1 x^n \ln(1+x) dx$ where $n = 0, 1, 2, \dots$ **3**
Show that

$$(n+1)I_n = 2 \ln 2 - \frac{1}{n+1} - nI_{n-1}$$

where $n = 1, 2, \dots$

- iii. Hence show that **4**

$$(n+1)I_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{n+1}$$

when n is odd.

- (b) A particle is moving in simple harmonic motion with period T about a centre O . Its displacement at any time t is given by $x = a \sin(nt)$, where a is the amplitude.

- i. Show that the velocity of the particle is **1**

$$\dot{x} = \frac{2a\pi}{T} \cos\left(\frac{2\pi}{T}t\right)$$

- ii. The point P lies D units on the positive side of O . Let V be the velocity of the particle when it first passes through P . **2**

Show that the first time the particle is at P after passing through O is

$$\frac{T}{2\pi} \tan^{-1}\left(\frac{2\pi D}{VT}\right)$$

- iii. If the second time the particle is at P after passing through O is $t = t_2$, **2**
show that

$$\tan\left(\frac{2\pi}{T}t_2\right) = -\frac{2\pi D}{VT}$$

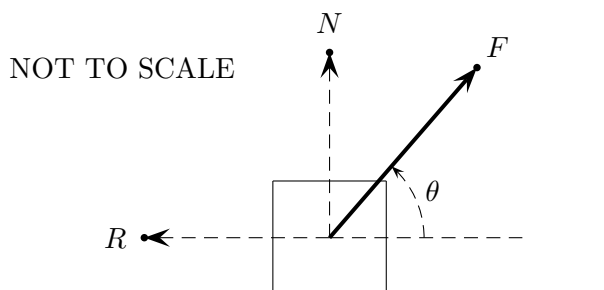
Examination continues overleaf...

Question 16 (13 Marks)

Commence a NEW booklet.

Marks

- (a) A block of mass 5 kg is to be moved along a rough horizontal surface by a force of magnitude F newtons, inclined at an angle of θ to the direction of motion, where $0 \leq \theta \leq \frac{\pi}{2}$.



There is a frictional force of magnitude R newtons, which is proportional to the normal reaction force of magnitude N newtons exerted on the block by the surface, such that $R = 0.2N$. Take $g = 10 \text{ ms}^{-2}$.

- i. Show that when the block is about to move,

3

$$F = \frac{50}{5 \cos \theta + \sin \theta} \text{ newtons}$$

- ii. Calculate the minimum value of F needed to overcome the frictional resistance between the block and the surface.

3

- (b) A projectile is launched from the origin with a velocity vector $\dot{\mathbf{r}} = \begin{pmatrix} u_0 \\ v_0 \end{pmatrix}$. It is subject to gravity and there is air resistance, which acts in the opposite direction to the instantaneous direction of motion. The magnitude of the air resistance is mkv , where v is the velocity of the projectile at any time t , and m is the mass of the projectile.

- i. Show that the position vector of the projectile is given by

5

$$\mathbf{r} = \begin{pmatrix} \frac{u_0}{k} (1 - e^{-kt}) \\ \left(\frac{g}{k^2} + \frac{v_0}{k} \right) (1 - e^{-kt}) - \frac{g}{k} t \end{pmatrix}$$

- ii. Hence, or otherwise, find the Cartesian equation of the path of the projectile.

2**End of paper.**

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Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

NESA STUDENT #:

Class (please ✓)

☐ 12MXX.1 – Mr Sekaran

☐ 12MXX.2 – Ms Ham

Directions for multiple choice answers

- Read each question and its suggested answers.
- Select the alternative (A), (B), (C), or (D) that best answers the question.
- Mark only one circle per question. There is only *one* correct choice per question.
- Fill in the response circle completely, using blue or black pen, e.g.

(A) (B) ● (D)

- If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) (B) ~~●~~ ●

- If you continue to change your mind, write the word **correct** and clearly indicate your final choice with an arrow as shown below:

(A) (B) ~~●~~ ~~●~~ ^{correct}

1 – (A) (B) (C) (D)

2 – (A) (B) (C) (D)

3 – (A) (B) (C) (D)

4 – (A) (B) (C) (D)

5 – (A) (B) (C) (D)

6 – (A) (B) (C) (D)

7 – (A) (B) (C) (D)

8 – (A) (B) (C) (D)

9 – (A) (B) (C) (D)

10 – (A) (B) (C) (D)

Sample Band E4 Responses

Section I

1. (D) 2. (C) or (D) 3. (B) 4. (C) 5. (A)
6. (C) 7. (C) 8. (D) 9. (D) 10. (D)

Section II

Question 11 ()

- (a) i. (2 marks)

✓ [1] for $p^3 = (2m+1)^3$.

✓ [1] for final conclusion with reasons

Assume p is odd. Then $\exists m \in \mathbb{Z}^+$ such that $p = 2m + 1$.

$$\begin{aligned} p^3 &= (2m+1)^3 \\ &= 8m^3 + 12m^2 + 6m + 1 \\ &= 2(4m^3 + 6m^2 + 3m) + 1 \\ &= 2N + 1, \text{ where } N = 4m^3 + 6m^2 + 3m \end{aligned}$$

such that $N \in \mathbb{Z}^+$. Hence p^3 is odd, which is a contradiction.

$\therefore$ If p^3 is even then p is even.

- ii. (3 marks)

✓ [1] for cubing to get $2 = \frac{p^3}{q^3}$

✓ [1] for deducing q is even

✓ [1] for final conclusion with reasons

Assume $\sqrt[3]{2}$ is rational.

$$\begin{aligned} \sqrt[3]{2} &= \frac{p}{q}, \text{ where } p \text{ and } q \text{ are co-prime} \\ 2 &= \frac{p^3}{q^3} \\ 2q^3 &= p^3 \end{aligned}$$

If p^3 is even, then p is even (proven in (i)).

Then $\exists k \in \mathbb{Z}$ such that $p = 2k$.

$$\begin{aligned} 2q^3 &= (2k)^3 \\ 2q^3 &= 8k^3 \\ q^3 &= 4k^3 \end{aligned}$$

$\therefore q^3$ is even, then q is even.

Since both p and q are even they are not co-prime, which is a contradiction.

$\therefore \sqrt[3]{2}$ is irrational.

- (b) (3 marks)

✓ [1] for explaining why $n = 3k + 1$

✓ [1] for getting $n^3 + (n+1)^3$

✓ [1] for final conclusion with reasons

Since neither n nor $n + 1$ is divisible by 3, $\exists k \in \mathbb{Z}^+$ such that $n = 3k + 1$ or $n = 3k + 2$.

$\therefore$ If $n = 3k + 2$ then $n + 1 = 3k + 3$, which is divisible by 3.

Thus n and $n + 1$ must be of the form $3k + 1$ and $3k + 2$ respectively.

$$\begin{aligned} n^3 + (n + 1)^3 &= [n + (n + 1)](n^2 - n(n + 1) + (n + 1)^2) \\ &= (2n + 1)(n^2 + n + 1) \\ &= (6k + 3)(6k^2 + 12k + 4 + 3k + 2 + 1) \\ &= (6k + 3)(6k^2 + 15k + 9) \\ &= 9(2k + 1)(2k^2 + 5k + 3) \end{aligned}$$

which is divisible by 9, since $2k + 1, 2k^2 + 5k + 3 \in \mathbb{Z}$.

(c) i. (4 marks)

✓ [1] for proving the base case.

✓ [2] for the inductive hypothesis, and for progress in using recurrence relation.

✓ [1] for final conclusion.

Let $P(n)$ be the proposition

- Base case: $P(1)$:

$$\begin{array}{ll} \text{LHS} & x_1 = 1 \\ \text{RHS} & 2 \left(\frac{1 + \left(-\frac{1}{3}\right)}{1 - \left(-\frac{1}{3}\right)} \right) = 2 \left(\frac{\frac{2}{3}}{\frac{4}{3}} \right) = 1 \end{array}$$

Hence $P(1)$ is true.

- Inductive step: assume $P(k)$ is true, $k \in \mathbb{Z}^+$:

$$x_k = 2 \left(\frac{1 + \alpha^k}{1 - \alpha^k} \right) \quad \forall k \in \mathbb{Z}^+$$

Prove $P(k+1)$ is true:

$$\begin{aligned}
 x_{k+1} &= \frac{4+x_k}{1+x_k} \\
 &= \frac{4+2\left(\frac{1+\alpha^k}{1-\alpha^k}\right)}{1+2\left(\frac{1+\alpha^k}{1-\alpha^k}\right)} \quad \dots \text{from the assumption} \\
 &= \frac{4-4\alpha^k+2+2\alpha^k}{1-\alpha^k+2+2\alpha^k} \\
 &= \frac{6-2\alpha^k}{3+\alpha^k} \\
 &= \frac{6\left(1-\frac{\alpha^k}{3}\right)}{3\left(1+\frac{\alpha^k}{3}\right)} \\
 &= \frac{2\left(1+\left(-\frac{1}{3}\right)\alpha^k\right)}{\left(1-\left(-\frac{1}{3}\right)\alpha^k\right)} \\
 &= 2\left(\frac{1+\alpha^{k+1}}{1-\alpha^{k+1}}\right)
 \end{aligned}$$

$\therefore P(k+1)$ is true.

By Mathematical induction, $P(n)$ is true for $n \in \mathbb{Z}^+$ where $n \geq 1$.

ii. (1 mark)

As $\alpha^n \rightarrow 0$ as $n \rightarrow \infty$, $x_n \rightarrow 2$.

(d) i. (2 marks)

✓ [1] for $ab \leq 9$

✓ [1] for showing the final result

If $a+b=6$,

$$\begin{aligned}
 \therefore a^2 + b^2 &\geq 2ab \\
 (a+b)^2 &\geq 4ab \\
 6^2 &\geq 4ab \\
 ab &\leq 9 \\
 \frac{1}{ab} &\geq \frac{1}{9} \\
 \therefore \frac{1}{a} + \frac{1}{b} &= \frac{a+b}{ab} \geq \frac{6}{9} = \frac{2}{3}
 \end{aligned}$$

ii. (2 marks)

✓ [1] for $\frac{1}{ab} \geq \frac{4}{c^2}$

✓ [1] for showing the final result

If $a + b = c$,

$$\begin{aligned}\therefore (a + b)^2 &\geq 4ab \\ c^2 &\geq 4ab \\ \frac{1}{ab} &\geq \frac{4}{c^2} \\ \therefore \frac{1}{a^2} + \frac{1}{b^2} &= \frac{a^2 + b^2}{(ab)^2} \\ &\geq \frac{2ab}{(ab)^2} \\ &= \frac{2}{ab} \\ &\geq 2 \left(\frac{4}{c^2} \right) \\ &= \frac{8}{c^2}\end{aligned}$$

Question 12 ()

(a) i. (1 mark)

$$\begin{aligned}
 \overrightarrow{AB} &= \begin{pmatrix} -5 \\ 4 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \\
 &= \begin{pmatrix} -6 \\ 6 \\ -4 \end{pmatrix} \\
 &= 2 \begin{pmatrix} -3 \\ 3 \\ -2 \end{pmatrix}
 \end{aligned}$$

$\therefore$ A vector equation is $\underline{r} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 3 \\ 2 \end{pmatrix}$, where $\lambda \in \mathbb{R}$.

ii. (3 marks)

✓ [1] for setting up three pairs of equations.

✓ [1] for values of λ and μ

✓ [1] for showing the lines are skew by substitution

$$\underline{r}_1 = \begin{pmatrix} 1 - 3\lambda \\ -2 + 3\lambda \\ 3 - 2\lambda \end{pmatrix} \text{ and } \underline{r}_2 = \begin{pmatrix} 1 - \mu \\ 2 + 3\mu \\ -1 + \mu \end{pmatrix}$$

$$\begin{cases} 1 - 3\lambda = 1 - \mu & \dots(1) \\ -2 + 3\lambda = 2 + 3\mu & \dots(2) \\ 3 - 2\lambda = 2 + 3\mu & \dots(3) \end{cases}$$

From (1), $\mu = 3\lambda$.

Sub this into (2),

$$-2 + 3\lambda = 2 + 9\lambda$$

$$6\lambda = -4$$

$$\lambda = -\frac{2}{3} \text{ and } \mu = -2$$

Sub these into (3),

$$\text{LHS} = 3 - 2\left(-\frac{2}{3}\right) = \frac{13}{3}$$

$$\text{RHS} = 2 + 3(-2) = -4$$

 $\therefore$ LHS $\neq$ RHS, the lines are skew.

(b) i. (1 mark)

$$\text{radius } r = \sqrt{6^2 + 2^2 + 4^2} = \sqrt{56} = 2\sqrt{14}$$

$$\therefore \text{The vector equation is } \left| \underline{u} - \begin{pmatrix} -3 \\ -5 \\ 10 \end{pmatrix} \right| = 2\sqrt{14}$$

ii. (1 mark)

$$(x+3)^2 + (x+5)^2 + (x-10)^2 = 56$$

iii. (2 marks)

✓ [1] for finding the distance between the two centres

✓ [1] for showing the final result

Distance between the two centres $(-3, -5, 10)$ and $(-9, 4, 7)$ is

$$\sqrt{6^2 + 9^2 + 3^2} = \sqrt{126} = 3\sqrt{14} = \sqrt{14} + 2\sqrt{14}$$

, which is the addition of two radii.

$\therefore S_1$ and S_2 touch each other at a single point.

iv. (3 marks)

✓ [1] for equating the two vector equations

✓ [1] for the quadratic equation

✓ [1] for two values of λ

$$\text{Equate } \underline{\mathbf{y}} = \begin{pmatrix} -6 + 2\lambda \\ -3 + \lambda \\ 11 + \lambda \end{pmatrix} \text{ and } \underline{\mathbf{u}} - \begin{pmatrix} -3 \\ -5 \\ 10 \end{pmatrix} = 2\sqrt{14}$$

$$\left| \begin{pmatrix} -6 + 2\lambda \\ -3 + \lambda \\ 11 + \lambda \end{pmatrix} - \begin{pmatrix} -3 \\ -5 \\ 10 \end{pmatrix} \right| = 2\sqrt{14}$$

$$\left| \begin{pmatrix} -3 + 2\lambda \\ 2 + \lambda \\ 1 + \lambda \end{pmatrix} \right| = 2\sqrt{14}$$

$$(-3 + 2\lambda)^2 + (2 + \lambda)^2 + (1 + \lambda)^2 = 2\sqrt{14}$$

$$9 - 12\lambda + 4\lambda^2 + 4 + 4\lambda + \lambda^2 + 1 + 2\lambda + \lambda^2 = 56$$

$$6\lambda^2 - 6\lambda - 42 = 0$$

$$\lambda^2 - \lambda - 7 = 0$$

$$\therefore \lambda = \frac{1 \pm \sqrt{29}}{2}$$

(c) i. (2 marks)

✓ [1] for $\overrightarrow{CE}$

✓ [1] for showing the final result

$$\overrightarrow{DC} = \frac{1}{3}\underline{\mathbf{c}} \text{ and } \overrightarrow{CB} = \underline{\mathbf{b}} - \underline{\mathbf{c}}$$

$$\overrightarrow{CE} = \frac{2}{3}(\underline{\mathbf{b}} - \underline{\mathbf{c}})$$

$$\therefore \overrightarrow{DE} = \overrightarrow{DC} + \overrightarrow{CE}$$

$$= \frac{1}{3}\underline{\mathbf{c}} + \frac{2}{3}(\underline{\mathbf{b}} - \underline{\mathbf{c}})$$

$$= \frac{2}{3}\underline{\mathbf{b}} - \frac{1}{3}\underline{\mathbf{c}}$$

ii. (3 marks)

✓ [1] for getting another expression of $\overrightarrow{AF}$

✓ [1] for values of μ and λ

✓ [1] for showing the final result

Let $\overrightarrow{AF} = \lambda \overrightarrow{AB} = \lambda \mathfrak{b}$ and $\overrightarrow{DF} = \mu \overrightarrow{DE} = \frac{\mu}{3}(2\mathfrak{b} - \mathfrak{c})$ where $\lambda, \mu \in \mathbb{R}$.

Also

$$\begin{aligned}\overrightarrow{AF} &= \overrightarrow{AD} + \overrightarrow{DF} = \frac{2}{3}\mathfrak{c} + \frac{\mu}{3}(2\mathfrak{b} - \mathfrak{c}) = \frac{2-\mu}{3}\mathfrak{c} + \frac{2\mu}{3}\mathfrak{b} \\ \therefore \frac{2-\mu}{3} &= 0 \text{ and } \frac{2\mu}{3} = \lambda \\ \mu &= 2 \text{ and } \lambda = \frac{4}{3}\end{aligned}$$

Sub $\lambda = \frac{4}{3}$ into $\overrightarrow{AF} = \lambda \overrightarrow{AB}$

$$\begin{aligned}3\overrightarrow{AF} &= 4\overrightarrow{AB} \\ \therefore 3AF &= 4AB = 4(AF - BF) \\ AF &= 4BF \\ \therefore AF : BF &= 4 : 1\end{aligned}$$

Question 13 (Ham)

(a) (3 marks)

✓ [1] for finding the third root

✓ [1] for value of p .

✓ [1] for value of q .

As $P(x)$ has real coefficients, then any complex roots that appear also have its conjugate appear as a root. Hence, $1+i$ and $1-i$ are roots of $z^3 + pz^2 + qz + 6 = 0$. Let the third root be α .

Product of roots:

$$\begin{aligned}(1+i)(1-i)\alpha &= -6 \\ 2\alpha &= -6 \\ \therefore \alpha &= -3\end{aligned}$$

Sum of roots,

$$\begin{aligned}(1+i) + (1-i) + (-3) &= -p \\ \therefore p &= 1\end{aligned}$$

Sum of product of two roots,

$$\begin{aligned}(1+i)(1-i) - 3(1+i) - 3(1-i) &= q \\ 2 - 6 &= q \\ \therefore q &= -4\end{aligned}$$

Hence roots are $1 \pm i$ and $1 \pm 2i$.

(b) i. (3 marks)

✓ [1] for correctly converting $1 + i$ and $1 - i$ to $e^{i\theta}$ form✓ [1] simplifying $e^{i\frac{\pi}{4}} + e^{-i\frac{\pi}{4}}$ to $2\cos\frac{n\pi}{4}$

✓ [1] for showing the final result

$$\begin{aligned}
 (1+i)^n + (1-i)^n &= \left(\sqrt{2}e^{i\frac{\pi}{4}}\right)^n + \left(\sqrt{2}e^{-i\frac{\pi}{4}}\right)^n \\
 &= \left(\sqrt{2}\right)^n \left(e^{i\frac{\pi}{4}} + e^{-i\frac{\pi}{4}}\right) \\
 &= \left(\sqrt{2}\right)^n \left(2\cos\frac{n\pi}{4}\right) \\
 &= \left(\sqrt{2}\right)^n \times \left(\sqrt{2}\right)^2 \cos\frac{n\pi}{4} \\
 &= \left(\sqrt{2}\right)^{n+2} \cos\frac{n\pi}{4}
 \end{aligned}$$

ii. (3 marks)

✓ [1] for expanding using binomial theorem.

✓ [1] for simplifying and equating with the expression in (i).

✓ [1] for showing the final result

$$\begin{aligned}
 (1+i)^n + (1-i)^n &= 1 + \binom{n}{1}i + \binom{n}{2}i^2 + \binom{n}{3}i^3 + \dots + \binom{n}{n}i^n + \\
 &\quad 1 - \binom{n}{1}i + \binom{n}{2}i^2 - \binom{n}{3}i^3 + \dots + \binom{n}{n}i^n \\
 &= 2 \left[\binom{n}{0} + \binom{n}{2}i^2 + \binom{n}{4}i^4 + \dots + \binom{n}{n}i^n \right] \\
 &= 2 \left[\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \dots + \binom{n}{n} \right] \\
 \therefore 2 \left[\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \dots + \binom{n}{n} \right] &= \left(\sqrt{2}\right)^{n+2} \cos\frac{n\pi}{4} \quad \dots \text{ using (i)} \\
 &= \left(\sqrt{2}\right)^{n+2} \cos(m\pi), \quad \text{where } n = 4m \\
 \therefore \binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \dots + \binom{n}{n} &= \left(\sqrt{2}\right)^n (-1)^m \\
 &= \left(\sqrt{2}\right)^n (-1)^{\frac{n}{4}}
 \end{aligned}$$

(c) i. (1 mark)

$$\mu^7 = \cos 2\pi + i \sin 2\pi = 1$$

$$\mu^7 - 1 = 0$$

$$(\mu - 1)(1 + \mu + \mu^2 + \dots + \mu^6) = 0$$

Since

$$\mu \neq 1, \quad 1 + \mu + \mu^2 + \dots + \mu^6 = 0$$

ii. (2 marks)

✓ [1] for $\beta = \bar{\mu} + \bar{\mu}^2 + \bar{\mu}^4$

✓ [1] for showing the final result.

Since α is a complex root, $\bar{\alpha}$ is also a root.

$$\beta = \bar{\alpha} = \overline{\mu + \mu^2 + \mu^4} = \bar{\mu} + \bar{\mu}^2 + \bar{\mu}^4$$

Since

$$\begin{aligned}\bar{\mu} &= \cos\left(-\frac{2\pi}{7}\right) + \sin\left(-\frac{2\pi}{7}\right) = \cos\frac{12\pi}{7} + \sin\frac{12\pi}{7} = \mu^6 \\ \bar{\mu}^2 &= \cos\left(-\frac{4\pi}{7}\right) + \sin\left(-\frac{4\pi}{7}\right) = \cos\frac{10\pi}{7} + \sin\frac{10\pi}{7} = \mu^5 \\ \bar{\mu}^4 &= \cos\left(-\frac{8\pi}{7}\right) + \sin\left(-\frac{8\pi}{7}\right) = \cos\frac{6\pi}{7} + \sin\frac{6\pi}{7} = \mu^3 \\ \therefore \beta &= \mu^3 + \mu^5 + \mu^6\end{aligned}$$

iii. (3 marks)

✓ [1] for value of a

✓ [1] for expanding and simplifying $\alpha\beta$

✓ [1] for value of b

Using sum of roots,

$$\begin{aligned}a &= -(\alpha + \beta) \\ &= -(\mu + \mu^2 + \mu^3 + \dots + \mu^6) \\ &= -(-1) \\ &= 1\end{aligned}$$

Using product of roots,

$$\begin{aligned}b &= \alpha\beta \\ &= (\mu + \mu^2 + \mu^4)(\mu^3 + \mu^5 + \mu^6) \\ &= \mu^4 + \mu^6 + \mu^7 + \mu^5 + \mu^7 + \mu^8 + \mu^7 + \mu^9 + \mu^{10} \\ &= \mu^4 + \mu^6 + 1 + \mu^5 + 1 + \mu + 1 + \mu^2 + \mu^3 \\ &= 3 + \mu + \mu^2 + \mu^3 + \dots + \mu^6 \\ &= 2\end{aligned}$$

iv. (3 marks)

✓ [1] for solving the quadratic equation

✓ [1] for evaluating α

✓ [1] for showing the final result

The quadratic equation is now $x^2 + x + 2 = 0$.

$$\begin{aligned}x &= \frac{-1 \pm \sqrt{1-8}}{2} \\ &= \frac{-1 \pm i\sqrt{7}}{2}\end{aligned}$$

Consider $\alpha = \mu + \mu^2 + \mu^4$,

$$\begin{aligned}\operatorname{Im}(\mu + \mu^2 + \mu^4) &= \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} \\ &= 1.3228... \\ &> 0 \\ \therefore \alpha &= \frac{-1 + \sqrt{7}i}{2}\end{aligned}$$

Now equate the imaginary parts

$$\begin{aligned}\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} &= \sin \frac{2\pi}{7} + \sin \left(\pi - \frac{3\pi}{7} \right) + \sin \left(\pi + \frac{\pi}{7} \right) \\ &= \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} \\ &= \frac{\sqrt{7}}{2}\end{aligned}$$

Question 14 ()

(a) (3 marks)

- ✓ [1] for values of A, B and C
- ✓ [1] for finding the primitive of $\int \frac{x+2}{x^2+1} dx$
- ✓ [1] for finding the primitive of $\int \frac{1}{x+2} dx$

$$\begin{aligned}\text{Let } \frac{4x+3}{(x^2+1)(x+2)} &= \frac{Ax+B}{x^2+1} + \frac{C}{x+2}, \quad \text{where } A, B, C \in \mathbb{R} \\ 4x+3 &= (Ax+B)(x+2) + C(x^2+1)\end{aligned}$$

Let $x = -2$

$$-5 = 5C, \quad C = -1$$

Let $x = 0$

$$3 = 2B + C, \quad B = 2$$

Compare the coefficients of x^2

$$0 = A + C, \quad A = 1$$

$$\begin{aligned}\therefore \int \frac{4x+3}{(x^2+1)(x+2)} dx &= \int \frac{x+2}{x^2+1} - \frac{1}{x+2} dx \\ &= \int \frac{x}{x^2+1} + \frac{2}{x^2+1} - \frac{1}{x+2} dx \\ &= \frac{1}{2} \ln(x^2+1) + 2 \tan^{-1} x - \ln|x+2| + C\end{aligned}$$

(b) i. (1 mark)

$$\begin{aligned}\sec^{2n} \theta &= (1 + \tan^2 \theta)^n \\ &= \sum_{k=0}^n \binom{n}{k} 1^{n-k} (\tan^2 \theta)^k \\ &= \sum_{k=0}^n \binom{n}{k} \tan^{2k} \theta\end{aligned}$$

ii. (2 marks)

✓ [1] for using (i) to change the integrand.

✓ [1] for final answer.

$$\begin{aligned}\int \sec^8 \theta \, d\theta &= \int \sec^6 \theta \sec^2 \theta \, d\theta \\ &= \int \left(\sum_{k=0}^3 \binom{3}{k} \tan^{2k} \theta \right) \sec^2 \theta \, d\theta \\ &= \sum_{k=0}^3 \binom{3}{k} \int \tan^{2k} \theta \sec^2 \theta \, d\theta \\ &= \sum_{k=0}^3 \binom{3}{k} \frac{1}{2k+1} \tan^{2k+1} \theta + C\end{aligned}$$

(c) i. (4 marks)

✓ [1] for transforming the differential.

✓ [1] for transforming both limits.

✓ [1] for transforming integrand to an integrable form.

✓ [1] for final answer.

$$\begin{aligned}x &= \sin \theta \\ \therefore dx &= \cos \theta \, d\theta\end{aligned}$$

Transforming the limits,

$$\begin{aligned}x &= \frac{\sqrt{3}}{2} & \theta &= \frac{\pi}{3} \\ x &= 0 & \theta &= 0\end{aligned}$$

$$\begin{aligned}
& \int_0^{\frac{\pi}{3}} \frac{\sin^3 \theta}{\cos \theta} \cos \theta \, d\theta \\
&= \int_0^{\frac{\pi}{3}} (1 - \cos^2 \theta) \sin \theta \, d\theta \\
&= \int_0^{\frac{\pi}{3}} \sin \theta - \sin \theta \cos^2 \theta \, d\theta \\
&= \left[-\cos \theta + \frac{\cos^3 \theta}{3} \right]_0^{\frac{\pi}{3}} \\
&= -\frac{1}{2} + \frac{1}{24} - \left(-1 + \frac{1}{3} \right) \\
&= \frac{5}{24}
\end{aligned}$$

ii. (2 marks)

✓ [1] for correctly integrating

✓ [1] for final answer

$$\begin{aligned}
I_n &= \int_0^{\frac{\sqrt{3}}{2}} 3x^2 \cos^{-1} x \, dx \\
&\left| \begin{array}{ll} u = \cos^{-1} x & v = x^3 \\ du = \frac{-1}{\sqrt{1-x^2}} & dv = 3x^2 \end{array} \right. \\
I_n &= \left[x^3 \cos^{-1} x \right]_0^{\frac{\sqrt{3}}{2}} + \int_0^{\frac{\sqrt{3}}{2}} \frac{x^3}{\sqrt{1-x^2}} \, dx \\
&= \left(\frac{3\sqrt{3}}{8} \times \frac{\pi}{6} - 0 \right) + \frac{5}{24} \quad \dots \text{from (i)} \\
&= \frac{\sqrt{3}\pi}{16} + \frac{5}{24}
\end{aligned}$$

Question 15 ()

(a) i. (2 marks)

✓ [1] for correct application of integration by parts

✓ [1] for final answer.

$$\begin{aligned}
& \int \ln(1+x) \, dx \\
&\left| \begin{array}{ll} u = \ln(1+x) & v = x \\ du = \frac{1}{1+x} & dv = 1 \end{array} \right. \\
\int \ln(1+x) \, dx &= x \ln(1+x) - \int \frac{x}{1+x} \, dx \\
&= x \ln(1+x) - \int 1 - \frac{1}{1+x} \, dx \\
&= x \ln(1+x) - x + \ln(1+x) + C
\end{aligned}$$

ii. (3 marks)

- ✓ [1] for correct application of integration by parts
- ✓ [1] for arriving at (*)
- ✓ [1] for showing the final result

$$\begin{aligned}
 I_n &= \int x^n \ln(1+x) dx \\
 &\left| \begin{array}{ll} u = x^n & v = (x+1) \ln(1+x) - x \\ du = nx^{n-1} & dv = \ln(1+x) \end{array} \right. \\
 I_n &= \left[x^n (x \ln(1+x) - x + \ln(1+x)) \right]_0^1 - n \int_0^1 x^{n-1} (x \ln(1+x) - x + \ln(1+x)) dx \\
 &= 2 \ln 2 - 1 - n \int_0^1 x^n \ln(1+x) + x^{n-1} \ln(1+x) - x^n dx \\
 &= 2 \ln 2 - 1 - n (I_n + I_{n-1}) + n \left[\frac{x^{n+1}}{n+1} \right]_0^1 \dots (*) \\
 &= 2 \ln 2 - 1 + \frac{n}{n+1} - nI_n - nI_{n-1} \\
 (n+1)I_n &= 2 \ln 2 + \frac{n-n-1}{n+1} - nI_{n-1} \\
 (n+1)I_n &= 2 \ln 2 - \frac{1}{n+1} - nI_{n-1}
 \end{aligned}$$

iii. (4 marks)

- ✓ [1] for an expression for I_{n-1}
- ✓ [1] for arriving at (15.1)
- ✓ [1] for calculating I_0
- ✓ [1] for showing the final result

$$\begin{aligned}
 (n+1)I_n &= 2 \ln 2 - \frac{1}{n+1} - nI_{n-1} \\
 &= 2 \ln 2 - \frac{1}{n+1} - \left(2 \ln 2 - \frac{1}{n} - (n-1)I_{n-2} \right) \\
 &= -\frac{1}{n+1} + \frac{1}{n} + (n-1)I_{n-2} \\
 &= -\frac{1}{n+1} + \frac{1}{n} + 2 \ln 2 - \frac{1}{n-1} - (n-2)I_{n-3} \\
 &= 2 \ln 2 - \frac{1}{n+1} + \frac{1}{n} - \frac{1}{n-1} - (n-2)I_{n-3}
 \end{aligned}$$

Since n is odd,

$$(n+1)I_n = 2 \ln 2 - \frac{1}{n+1} + \frac{1}{n} - \frac{1}{n-1} + \dots + \frac{1}{3} - \frac{1}{2} - I_0 \quad (15.1)$$

$$\begin{aligned} \therefore I_0 &= \int_0^1 \ln(1+x) dx \\ &= [(x+1) \ln x - x]_0^1 \\ &= 2 \ln 2 - 1 \end{aligned}$$

$$\begin{aligned} \therefore (n+1)I_n &= \cancel{2 \ln 2} - \frac{1}{n+1} + \frac{1}{n} - \frac{1}{n-1} + \dots + \frac{1}{3} - \frac{1}{2} - (\cancel{2 \ln 2} - 1) \\ &= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{n+1} \end{aligned}$$

(b) i. (1 mark)

$$\begin{aligned} x &= a \sin(nt) \\ \dot{x} &= an \cos(nt) \end{aligned}$$

Since $\frac{2\pi}{n} = T$, $n = \frac{2\pi}{T}$

$$\therefore \dot{x} = \frac{2a\pi}{T} \cos\left(\frac{2\pi}{T}t\right)$$

ii. (2 marks)

✓ [1] for equations (1) and (2)

✓ [1] for showing the final result

When $t = t_1$, $x = D$ and $\dot{x} = V$.

$$D = a \sin(nt_1) \quad \dots(1)$$

$$V = an \cos(nt_1) \quad \dots(2)$$

$$(1) \div (2),$$

$$\begin{aligned} \frac{D}{V} &= \frac{1}{n} \tan(nt_1) \\ \therefore t_1 &= \frac{1}{n} \tan^{-1}\left(\frac{nD}{V}\right) \\ &= \frac{T}{2\pi} \tan^{-1}\left(\frac{2\pi D}{VT}\right) \end{aligned}$$

iii. (2 marks)

✓ [1] for getting an expression for $-V$

✓ [1] for getting the final result

Let the particle comes back to P at $t = t_2$. Then,

$$D = a \sin\left(\frac{2\pi}{T}t_2\right) \quad \dots(1)$$

$$-V = \frac{2a\pi}{T} \cos\left(\frac{2\pi}{T}t_2\right) \quad \dots(2)$$

$$(1) \div (2),$$

$$-\frac{D}{V} = \frac{T}{2\pi} \tan\left(\frac{2\pi}{T}t_2\right)$$

$$\therefore \tan\left(\frac{2\pi}{T}t_2\right) = -\frac{2\pi D}{VT}$$

Question 16 ()

(a) i. (3 marks)

✓ [1] for (1)

✓ [1] for (2)

✓ [1] for showing the final result

Horizontal components:

$$F \cos \theta - 0.2N = 0 \quad \dots(1)$$

Vertical components:

$$N + F \sin \theta - 5g = 0 \quad \dots(2)$$

$$(1) + 0.2 \times (2),$$

$$F \cos \theta + 0.2F \sin \theta = 5g \times 0.2$$

$$F(\cos \theta + 0.2 \sin \theta) = 10$$

$$F = \frac{10}{\cos \theta + \frac{1}{5} \sin \theta}$$

$$= \frac{50}{5 \cos \theta + \sin \theta}$$

ii. (3 marks)

✓ [1] for arriving at min F when $5 \cos \theta + \sin \theta$ is max

✓ [1] for value of R

✓ [1] for final answer

Minimum F when $5 \cos \theta + \sin \theta$ is maximum.

Let $5 \cos \theta + \sin \theta = R \cos(\theta - \alpha)$.

$$5 \cos \theta + \sin \theta = R \cos \alpha \cos \theta + R \sin \alpha \sin \theta$$

$$R \cos \alpha = 5, \quad R \sin \alpha = 1$$

$$\therefore R = \sqrt{26}$$

$$\text{Now } 5 \cos \theta + \sin \theta = \sqrt{26} (\cos(\theta - \alpha))$$

$$\Rightarrow \text{The maximum value of } 5 \cos \theta + \sin \theta \text{ is } \sqrt{26}$$

$$\therefore \text{Min } F = \frac{50}{\sqrt{26}}$$

(b) i. (5 marks)

✓ [1] for an expression for v_x

- ✓ [1] for an expression for x
- ✓ [1] for an expression for $\ddot{y}$
- ✓ [1] for an expression for v_y
- ✓ [1] for an expression for y

Horizontal component:

$$\begin{aligned}
 m\ddot{x} &= -mkv_x \\
 \frac{dv_x}{dt} &= -kv_x \\
 \int \frac{1}{v_x} dv_x &= \int -k dt \\
 \ln |v_x| &= -kt + C_1
 \end{aligned}$$

$\therefore$ when $t = 0$, $v_x = u_0$

$$\begin{aligned}
 C_1 &= \ln u_0 \\
 \therefore \ln(v_x) &= -kt + \ln u_0 \\
 \ln\left(\frac{v_x}{u_0}\right) &= -kt \\
 v_x &= u_0 e^{-kt}
 \end{aligned}$$

Now

$$\begin{aligned}
 \frac{dx}{dt} &= u_0 e^{-kt} \\
 x &= u_0 \int e^{-kt} dt \\
 &= -\frac{u_0}{k} e^{-kt} + C_2
 \end{aligned}$$

$\therefore$ when $t = 0$, $x = 0$

$$\begin{aligned}
 C_2 &= \frac{u_0}{k} \\
 \therefore x &= -\frac{u_0}{k} e^{-kt} + \frac{u_0}{k} \\
 &= \frac{u_0}{k} (1 - e^{-kt})
 \end{aligned}$$

Vertical component:

$$\begin{aligned}
 m\ddot{y} &= -mg - mkv_y \\
 \frac{dv_y}{dt} &= -g - kv_y \\
 \int \frac{1}{g + kv_y} dv_y &= -\int dt \\
 \frac{1}{k} \ln |g + kv_y| &= -t + C_3
 \end{aligned}$$

$\therefore$ when $t = 0, v_y = v_0$

$$\begin{aligned} C_3 &= \frac{1}{k} \ln |g + kv_y| \\ \therefore t &= -\frac{1}{k} \ln \left| \frac{g + kv_y}{g + kv_0} \right| \\ \frac{g + kv_y}{g + kv_0} &= e^{-kt} \\ g + kv_y &= (g + kv_0)e^{-kt} \\ v_y &= \frac{g + kv_0}{k} e^{-kt} - \frac{g}{k} \end{aligned}$$

Now

$$\begin{aligned} \frac{dy}{dt} &= \left(\frac{g}{k} + v_0 \right) e^{-kt} - \frac{g}{k} \\ y &= -\left(\frac{g}{k^2} + \frac{v_0}{k} \right) e^{-kt} - \frac{g}{k} t + C_4 \end{aligned}$$

$\therefore$ when $t = 0, y = 0$

$$\begin{aligned} C_4 &= \frac{g}{k^2} + \frac{v_0}{k} \\ \therefore y &= -\left(\frac{g}{k^2} + \frac{v_0}{k} \right) e^{-kt} - \frac{g}{k} t + \left(\frac{g}{k^2} + \frac{v_0}{k} \right) \\ &= \left(\frac{g}{k^2} + \frac{v_0}{k} \right) (1 - e^{-kt}) - \frac{g}{k} t \\ \therefore \tilde{x} &= \left(\left(\frac{g}{k^2} + \frac{v_0}{k} \right) (1 - e^{-kt}) - \frac{g}{k} t \right) \end{aligned}$$

ii. (2 marks)

✓ [1] for an expression for t

✓ [1] for the final answer

From $x = \frac{u_0}{k} (1 - e^{-kt})$,

$$\begin{aligned} 1 - e^{-kt} &= \frac{kx}{u_0} \\ e^{-kt} &= 1 - \frac{kx}{u_0} \\ t &= -\frac{1}{k} \ln \left(1 - \frac{kx}{u_0} \right) \end{aligned}$$

Sub this into $y = \left(\frac{g}{k^2} + \frac{v_0}{k} \right) (1 - e^{-kt}) - \frac{g}{k} t$

$$\begin{aligned} y &= \left(\frac{g}{k^2} + \frac{v_0}{k} \right) \times \frac{kx}{u_0} + \frac{g}{k} \times \frac{1}{k} \ln \left(1 - \frac{kx}{u_0} \right) \\ \therefore y &= \frac{x}{u_0} \left(\frac{g}{k} + v_0 \right) + \frac{g}{k^2} \ln \left(1 - \frac{kx}{u_0} \right) \end{aligned}$$